PROGRAM 1:

```
GENERATE    18,6 ;
      QUEUE      Line ;
      SEIZE      Barber ;
      DEPART     Line  ;
      ADVANCE    16,4  ;
      RELEASE    Barber ;
      TERMINATE          ;
      GENERATE   480 ;
      TERMINATE  1   ;
      START      1   ;
```

OUTPUT:

GPSS World Simulation Report - Barber shop modeling.7.1

Thursday, February 05, 2026 11:09:28

START TIME	END TIME	BLOCKS	FACILITIES	STORAGES
0.000	480.000	9	1	0

NAME	VALUE
BARBER	10001.000
LINE	10000.000

LABEL	LOC	BLOCK TYPE	ENTRY COUNT	CURRENT COUNT	RETRY
	1	GENERATE	26	0	0
	2	QUEUE	26	0	0
	3	SEIZE	26	0	0
	4	DEPART	26	0	0
	5	ADVANCE	26	1	0
	6	RELEASE	25	0	0
	7	TERMINATE	25	0	0
	8	GENERATE	1	0	0
	9	TERMINATE	1	0	0

FACILITY	ENTRIES	UTIL.	AVE. TIME	AVAIL.	OWNER	PEND	INTER	RETRY	DELAY
BARBER	26	0.863	15.926	1	27	0	0	0	0

QUEUE	MAX	CONT.	ENTRY	ENTRY(0)	AVE.CONT.	AVE.TIME	AVE.(-0)	RETRY
LINE	1	0	26	15	0.078	1.441	3.407	0

FEC XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
27	0	482.664	27	5	6		
28	0	482.707	28	0	1		
29	0	960.000	29	0	8		

PROGRAM 2

```
; Time unit: 1 second


; Category 1 Mechanic
    GENERATE    420,360,,,1     ; Category 1 Mechanics arrive
    QUEUE       Line            ; Enter "Category 1 Segment" of line
    SEIZE       Clerk           ; Capture the clerk
    DEPART      Line            ; Leave the line
    ADVANCE     300,90          ; Use the clerk
    RELEASE     Clerk           ; Free the clerk
    TERMINATE                   ; Leave the tool crib area


; Category 2 Mechanic
    GENERATE    360,240,,,2     ; Category 2 Mechanics arrive
    QUEUE       Line            ; Enter "Category 2 Segment" of line
    SEIZE       Clerk           ; Capture the clerk
    DEPART      Line            ; Leave the line
    ADVANCE     100,30          ; Use the clerk
    RELEASE     Clerk           ; Free the clerk
    TERMINATE                   ; Leave the tool crib area


; Timer
    GENERATE    28800           ; Timer arrives after 8 hours
    TERMINATE   1               ; Shut off the run


; Control
```

```
START      1            ; Start the run
```

OUTPUT:

GPSS World Simulation Report - Untitled Model 6.2.1

Thursday, February 05, 2026 11:14:12

START TIME	END TIME	BLOCKS	FACILITIES	STORAGES
0.000	28800.000	16	1	0

NAME	VALUE
CLERK	10001.000
LINE	10000.000

LABEL	LOC	BLOCK TYPE	ENTRY COUNT	CURRENT COUNT	RETRY
	1	GENERATE	71	0	0
	2	QUEUE	71	6	0
	3	SEIZE	65	0	0
	4	DEPART	65	0	0
	5	ADVANCE	65	1	0
	6	RELEASE	64	0	0
	7	TERMINATE	64	0	0
	8	GENERATE	83	0	0

9	QUEUE	83	2	0
10	SEIZE	81	0	0
11	DEPART	81	0	0
12	ADVANCE	81	0	0
13	RELEASE	81	0	0
14	TERMINATE	81	0	0
15	GENERATE	1	0	0
16	TERMINATE	1	0	0

FACILITY	ENTRIES	UTIL.	AVE. TIME	AVAIL.	OWNER	PEND	INTER	RETRY	DELAY
CLERK	146	0.967	190.733	1	141	0	0	0	8

QUEUE	MAX	CONT.	ENTRY	ENTRY(0)	AVE.CONT.	AVE.TIME	AVE.(-0)	RETRY
LINE	9	8	154	6	2.616	489.248	509.082	0

FEC XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
141	1	28815.063	141	5	6		
157	2	29012.031	157	0	8		
155	1	29012.150	155	0	1		
158	0	57600.000	158	0	15		

- **gpss.py**

- GitHub

Examples

All examples on this page can be found in gpss.py's GitHub repository.

Contents

- Barber Shop
- Tool Crib
- Widget Assembly Line
- Inspection Station

Barber Shop

A one-line, one-server queuing system. Adapted from Case Study 2A of Thomas J. Schriber's "A GPSS Primer" (page II-36).

Program

```
; Time unit: 1 minute


; Customer
    GENERATE    18,6      ; Customers arrive
    QUEUE       Line      ; Enter the line
    SEIZE       Barber    ; Capture the barber
    DEPART      Line      ; Leave the line
    ADVANCE     16,4      ; Use the barber
    RELEASE     Barber    ; Free the barber
    TERMINATE             ; Leave the shop


; Timer
```

```
        GENERATE    480        ; Timer arrives at time 480 (8 hours)
        TERMINATE   1          ; Shut off the run

; Control
        START       1          ; Start the run
        END                    ; Exit the program
```

Output

gpss.py Simulation Report - examples/barber.gps

Generated on Saturday, June 05, 2021 at 17:33:34

```
============================ SIMULATION 1 ============================

Relative Clock: 480
Absolute Clock: 480

Facilities: 1

  "Barber":
    Avg. utilization: 86.04%
    Entries: 26
    Avg. time/Trans.: 15.885
    Available: no

Queues: 1

  "Line":
```

Maximum content: 1

Average content: 0.160

Total entries: 27

Zero entries: 12

Percent zeros: 44.44%

Avg. time/Trans.: 2.852

$ Avg. time/Trans.: 5.133

Current content: 1

Storages: 0

Tool Crib

A one-line, one-server queuing system with 2 customer types and priority distinctions. Adapted from Case Study 2C of Thomas J. Schriber's "A GPSS Primer" (page II-84).

Program

```
; Time unit: 1 second


; Category 1 Mechanic

    GENERATE    420,360,,,1     ; Category 1 Mechanics arrive

    QUEUE       Line            ; Enter "Category 1 Segment" of line

    SEIZE       Clerk           ; Capture the clerk

    DEPART      Line            ; Leave the line

    ADVANCE     300,90          ; Use the clerk

    RELEASE     Clerk           ; Free the clerk

    TERMINATE                   ; Leave the tool crib area


; Category 2 Mechanic
```

```
    GENERATE    360,240,,,2     ; Category 2 Mechanics arrive
    QUEUE       Line            ; Enter "Category 2 Segment" of line
    SEIZE       Clerk           ; Capture the clerk
    DEPART      Line            ; Leave the line
    ADVANCE     100,30          ; Use the clerk
    RELEASE     Clerk           ; Free the clerk
    TERMINATE                   ; Leave the tool crib area

; Timer
    GENERATE    28800           ; Timer arrives after 8 hours
    TERMINATE   1               ; Shut off the run

; Control
    START       1               ; Start the run
    END                         ; Exit the program
```

Output

gpss.py Simulation Report - examples/tool-crib.gps

Generated on Saturday, June 05, 2021 at 17:33:34

========== SIMULATION 1 ==========

Relative Clock: 28800

Absolute Clock: 28800

Facilities: 1

"Clerk":

Avg. utilization: 92.58%

Entries: 147

Avg. time/Trans.: 181.374

Available: no

Queues: 1

"Line":

Maximum content: 5

Average content: 1.294

Total entries: 149

Zero entries: 17

Percent zeros: 11.41%

Avg. time/Trans.: 250.040

$ Avg. time/Trans.: 282.242

Current content: 2

Storages: 0

Widget Assembly Line

A one-line, one-server queuing system with feedback. Adapted from Case Study 2D of Thomas J. Schriber's "A GPSS Primer" (page II-99).

Program

; Time unit: 1 minute

; Assembler

```
Key:    GENERATE    ,,,4    ; Provide 4 assemblers
Back:
        ADVANCE     30,5    ; Assemble next widget
        SEIZE       Oven    ; Capture the oven
        ADVANCE     8,2     ; Use the oven
        RELEASE     Oven    ; Free the oven
        TRANSFER    ,Back   ; Go do the next assembly

; Timer
        GENERATE    2400    ; Timer arrives after 5 days
        TERMINATE   1       ; Shut off the run

; Control and Block redefinitions
        START       1       ; Start the 1st run
Key:    GENERATE    ,,,5    ; Reconfigure for 2nd run
        CLEAR               ; Clear for 2nd run
        START       1       ; Start the 2nd run
Key:    GENERATE    ,,,6    ; Reconfigure for 3rd run
        CLEAR               ; Clear for 3rd run
        START       1       ; Start the 3rd run
        END                 ; Exit the program
```

Output

```
gpss.py Simulation Report - examples/widgets.gps
Generated on Saturday, June 05, 2021 at 17:33:34

============================ SIMULATION 1 ============================
```

Relative Clock: 2400

Absolute Clock: 2400

Facilities: 1

"Oven":

Avg. utilization: 78.88%

Entries: 237

Avg. time/Trans.: 7.987

Available: no

Queues: 0

Storages: 0

========== SIMULATION 2 ==========

Relative Clock: 2400

Absolute Clock: 2400

Facilities: 1

"Oven":

Avg. utilization: 94.04%

Entries: 285

Avg. time/Trans.: 7.919

Available: no

Queues: 0

Storages: 0

========== SIMULATION 3 ==========

Relative Clock: 2400

Absolute Clock: 2400

Facilities: 1

"Oven":

Avg. utilization: 98.92%

Entries: 295

Avg. time/Trans.: 8.047

Available: no

Queues: 0

Storages: 0

Inspection Station

Adapted from Case Study 2F of Thomas J. Schriber's "A GPSS Primer" (page II-144).

Program

```
; Time unit: 0.1 minutes

; Storage capacity definition
Inspection: STORAGE 2

; Television set
    GENERATE    55,20          ; Sets arrive from preceding station
Inspect:
    QUEUE       InspectWait    ; Enter inspection waiting area
    ENTER       Inspection     ; Capture an inspector
    DEPART      InspectWait    ; Leave the waiting area
    ADVANCE     90,30          ; Set is being inspected
    LEAVE       Inspection     ; Free the inspector
    TRANSFER    .15,,Adjust    ; Proceed to packing or adjustment station
    TERMINATE                  ; Set moves on to packing
Adjust:
    QUEUE       AdjustWait     ; Enter adjustment waiting area
    SEIZE       Adjustor       ; Capture the adjustor
    DEPART      AdjustWait     ; Leave the waiting area
    ADVANCE     300,100        ; Set is being adjusted
    RELEASE     Adjustor       ; Free the adjustor
    TRANSFER    ,Inspect       ; Go back to be inspected

; Timer
    GENERATE    4800           ; Timer arrives at end of each day
    TERMINATE   1              ; Provide snap output or shut off the run
```

```
; Control
        START      5,,1          ; Start the run
        END                      ; Exit the program
```

Output

```
gpss.py Simulation Report - examples/inspection.gps
Generated on Saturday, June 05, 2021 at 17:33:34

============================ SIMULATION 1 ============================

Relative Clock: 4800
Absolute Clock: 4800

Facilities: 1

  "Adjustor":
    Avg. utilization: 55.54%
    Entries: 9
    Avg. time/Trans.: 296.222
    Available: yes

Queues: 2

  "InspectWait":
    Maximum content: 2
    Average content: 0.281
```

Total entries: 93

Zero entries: 50

Percent zeros: 53.76%

Avg. time/Trans.: 14.505

$ Avg. time/Trans.: 31.372

Current content: 0

"AdjustWait":

Maximum content: 2

Average content: 0.281

Total entries: 9

Zero entries: 4

Percent zeros: 44.44%

Avg. time/Trans.: 150.000

$ Avg. time/Trans.: 270.000

Current content: 0

Storages: 1

"Inspection":

Capacity: 2

Average content: 1.676

Avg. utilization: 83.82%

Entries: 93

Avg. time/Trans.: 86.527

Maximum content: 2

Current content: 2

Remaining: 0

Available: no

========== SIMULATION 1 ==========

Relative Clock: 9600

Absolute Clock: 9600

Facilities: 1

"Adjustor":

Avg. utilization: 60.72%

Entries: 20

Avg. time/Trans.: 291.450

Available: no

Queues: 2

"InspectWait":

Maximum content: 3

Average content: 0.358

Total entries: 191

Zero entries: 84

Percent zeros: 43.98%

Avg. time/Trans.: 17.984

$ Avg. time/Trans.: 32.103

Current content: 2

"AdjustWait":

Maximum content: 2

Average content: 0.159

Total entries: 20

Zero entries: 12

Percent zeros: 60.00%

Avg. time/Trans.: 76.100

$ Avg. time/Trans.: 190.250

Current content: 0

Storages: 1

"Inspection":

Capacity: 2

Average content: 1.769

Avg. utilization: 88.47%

Entries: 189

Avg. time/Trans.: 89.878

Maximum content: 2

Current content: 2

Remaining: 0

Available: no

========== SIMULATION 1 ==========

Relative Clock: 14400

Absolute Clock: 14400

Facilities: 1

"Adjustor":

Avg. utilization: 74.72%

Entries: 37

Avg. time/Trans.: 290.784

Available: no

Queues: 2

"InspectWait":

Maximum content: 3

Average content: 0.487

Total entries: 293

Zero entries: 106

Percent zeros: 36.18%

Avg. time/Trans.: 23.915

$ Avg. time/Trans.: 37.471

Current content: 2

"AdjustWait":

Maximum content: 3

Average content: 0.493

Total entries: 37

Zero entries: 14

Percent zeros: 37.84%

Avg. time/Trans.: 191.946

$ Avg. time/Trans.: 308.783

Current content: 0

Storages: 1

"Inspection":

Capacity: 2

Average content: 1.813

Avg. utilization: 90.67%

Entries: 291

Avg. time/Trans.: 89.739

Maximum content: 2

Current content: 2

Remaining: 0

Available: no

========== SIMULATION 1 ==========

Relative Clock: 19200

Absolute Clock: 19200

Facilities: 1

"Adjustor":

Avg. utilization: 76.21%

Entries: 50

Avg. time/Trans.: 292.660

Available: no

Queues: 2

"InspectWait":

Maximum content: 4

Average content: 0.571

Total entries: 394

Zero entries: 125

Percent zeros: 31.73%

Avg. time/Trans.: 27.827

$ Avg. time/Trans.: 40.758

Current content: 1

"AdjustWait":

Maximum content: 3

Average content: 0.449

Total entries: 52

Zero entries: 18

Percent zeros: 34.62%

Avg. time/Trans.: 165.750

$ Avg. time/Trans.: 253.500

Current content: 2

Storages: 1

"Inspection":

Capacity: 2

Average content: 1.839

Avg. utilization: 91.94%

Entries: 393

Avg. time/Trans.: 89.837

Maximum content: 2

Current content: 2

Remaining: 0

Available: no

========== SIMULATION 1 ==========

Relative Clock: 24000

Absolute Clock: 24000

Facilities: 1

"Adjustor":

Avg. utilization: 78.84%

Entries: 65

Avg. time/Trans.: 291.108

Available: no

Queues: 2

"InspectWait":

Maximum content: 4

Average content: 0.598

Total entries: 496

Zero entries: 149

Percent zeros: 30.04%

Avg. time/Trans.: 28.938

$ Avg. time/Trans.: 41.363

Current content: 0

"AdjustWait":

Maximum content: 3

Average content: 0.535

Total entries: 67

Zero entries: 19

Percent zeros: 28.36%

Avg. time/Trans.: 191.746

$ Avg. time/Trans.: 267.646

Current content: 2

Storages: 1

"Inspection":

Capacity: 2

Average content: 1.849

Avg. utilization: 92.44%

Entries: 496

Avg. time/Trans.: 89.458

Maximum content: 2

Current content: 2

Remaining: 0

Available: no